	Class	Index No.
Candidate Name: ______________________________		

millennia institute

Promotional Examination 2008

Pre-University 2

MATHEMATICS
Higher 1 **8863/01**

WEDNESDAY 10 SEPTEMBER 3 hours

Additional materials:
Cover Page
Writing papers
List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class in the spaces at the top of this page and on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

This question paper consists of **6** printed pages.

[Turn over]

Section A: Pure Mathematics [40 marks]

1. Solve each of the following equations:

(i) $\log_3(2x+1)-\log_3(x-7)=2$, [3]

(ii) $\dfrac{8^{2x}}{4^{x+1}}=2^{2x+1}$. [3]

2. (a) On a single diagram, sketch the graphs of $y=|2x-3|$ and $y=x^2$. Hence, solve the inequality $|2x-3|<x^2$. [4]

(b) Find the values of m for which the line $y=mx-2$ is a tangent to the curve $y=\dfrac{1}{2}x^2$. [4]

3. Sketch the graph of $y=\dfrac{6x+1}{2x+1}$, clearly indicating the equations of asymptotes, and axial intercepts. [3]

State the gradient of the tangent to the curve at the point (0, 1). [1]

4. Functions f and g are defined by

$f:x\mapsto(x-1)^2,\ x\geq 1$,

$g:x\mapsto e^{-x},\ x\in\Re$.

(i) Find g^{-1}. [3]

(ii) Explain why fg does not exist. [2]

(iii) Given that the domain of g is restricted to the set $\{x\in\Re : x\leq k\}$, state the maximum value of k for which fg exists. [1]

5. (a) The equation of a curve is given by $y = x^5 + kx^4 + 3$, where k is a positive integer. Show that one of its stationary points occurs at $x = -\frac{4}{5}k$. [3]

Determine the nature of the other stationary point. [2]

(b) Differentiate $\ln\left(\frac{e^{2x}}{x+2}\right)$ with respect to *x*. [3]

6. (a) Sketch the graph of $y = 3x(x-2)$. Find the area enclosed by the curve, the *x*-axis, the lines x = 1 and x = 3. [3]

(b) Find

(i) $\int_{-3}^{5} 2^x \, dx$, [1]

(ii) $\int \frac{1}{(2-x)^3} dx$, [2]

(iii) $\int \frac{5x^3+1}{x} dx$. [2]

[Turn over]

Section B: Statistics [60 marks]

7. (a) Millennia Institute has 3 photocopiers namely A, B and C for printing papers. A large number of papers were printed. It was found that 50% of papers were printed from A, 30% from B, and 20% from C. Records showed that the occurrences of misaligned papers from printers A, B and C are 2%, 3% and 4% respectively. A printed sheet is chosen at random.

   (i) Construct a tree diagram and find the probability that the selected paper is misaligned. [3]

   (ii) Given that the selected paper is misaligned, find the probability that it was printed from A. [2]

   (b) The events A and B are mutually exclusive, while the events A and C are independent. Given that $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{10}$ and $P(A \cup C) = \frac{7}{10}$, find

   (i) $P(A \cup B)$, [2]

   (ii) $P(A \cap C)$. [3]

8. (a) In a binomial model, there are n independent trials and the probability of success for each trial is p. If the mean is 12 and the variance is 3, find the value of n and of p. [3]

   (b) In a certain college, the probability that a randomly selected student will enrol in an overseas university is 0.3. From a sample of 10 students chosen at random, find the probability that

   (i) only 3 will enrol in an overseas university, [2]

   (ii) less than 5 will enrol in an overseas university. [2]

   Each cohort has an average enrolment of 900 students. Use a suitable approximation to estimate the probability that at least 300 students of the cohort will enrol in an overseas university. [4]

9. After an extended period of monitoring, the mean time that Ah Boon takes to travel to school from home is found to be 24.5 minutes. Mr. Lim, his home tutor wishes to help him reduce his traveling time and recommended an alternative bus service. Ah Boon's traveling time using this alternative bus service, t minutes, on 72 randomly chosen mornings was noted and it was found that $\sum(t-20)=215$ and $\sum(t-20)^2=3234$.

(i) Find the unbiased estimates of the population mean and variance of the traveling time by the alternative bus service. [3]

(ii) Test at 5% level of significance, whether Ah Boon's traveling time from home to school is reduced by taking the alternative bus service. [4]

(iii) At 1% level of significance, calculate the minimum number of mornings that Ah Boon needs to monitor his traveling time to conclude that his traveling time is indeed reduced. [3]

10. (a) Calculate the product moment correlation coefficient between x and y given the following summarized data from a sample of 6 pairs of observations:

$\sum x = 24.3$, $\sum x^2 = 164.75$, $\sum y = 196.0$, $\sum y^2 = 18080.06$, $\sum xy = 667.07$. [3]

(b) The Mathematics test marks, x, and Geography test marks, y, of 5 students who sat for both tests are tabulated as follow:

x	56	41	75	88	34
y	32	24	70	65	47

Draw a scatter diagram for the above data set. [2]

Determine the equation of the regression lines of

(i) y on x, [1]

(ii) x on y. [1]

A student, who missed the Mathematics test, scored 55 for the Geography test. Estimate, using an appropriate regression line, the Mathematics mark of the student. Comment on the reliability of the result. [3]

[Turn Over]

11. (a) Give a reason why it is often more feasible to take samples instead of the entire population in a statistical study. [1]

(b) Give 1 advantage and 1 disadvantage associated with systematic sampling. [2]

(c) A study found that only 60% of the student population patronises Millennia Institute's new café. The management wishes to obtain the views of 30 students on the café.

(i) Describe how you would select the interviewees using simple random sampling. [2]

(ii) Describe an alternative sampling method if the management wants the interviewees to be a good representation of the student population. [2]

12. A large number of candidates sat for both a Mathematics and Economics examination. The marks obtained in the Mathematics and Economics examinations may be considered to be independent and normally distributed with means 43 and 60, and variances 20 and 35 respectively.

(i) Find the probability that a candidate scores at least 50 marks for Mathematics. [1]

(ii) Find the probability that a candidate's Mathematics mark exceeds half of his Economics mark by between 6 and 10 marks. [3]

(iii) If the probability that the mean Mathematics mark of n candidates exceeding 45 marks is less than 0.01, find the least value of n. [4]

(iv) A candidate is deemed to have passed if he scores at least 45 marks for any subject. Find the probability that a candidate passes both subjects, given that he scored a total of at least 90 marks for both Mathematics and Economics. [4]

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